

Design and Analysis of Braking System for Formula Student Car

Abstract:- This paper discusses the components used for the brake system and analysis of brake rotor. A brake is used for slowing or stopping a moving vehicle, typically by applying pressure to wheels. The material used for the rotor is mostly stainless steel and sometimes grey cast iron, carbon-carbon composite in high-end cars or bikes. In addition, it is necessary to study stress distribution, heat generation to improve braking efficiency. The rotor is created in Solid Works and imported into ANSYS for analysis. Trident-racing team implemented this braking system for Supra 19 event organized by SAE India.

INTRODUCTION

The Society of Automotive Engineers (FSAE) is an interdisciplinary competition organized by the Society of Automotive Engineers (SAE). It challenges university students (graduate and undergraduate students) from all over the world and provides them with their own car design, construction and competition. High performance, durability and reliability must be the attributes of this car. The car is not only tested under dynamic racing conditions, but also based on its design, functionality, marketability and cost. The design aims to ensure an effective and efficient braking system. A brake is a device that is applied to a moving machine part through artificial frictional resistance to stop the machine's movement, thereby stopping the machine's movement. In the process of performing this function, the brake absorbs the kinetic energy of moving parts or the potential energy released by descending objects such as elevators and elevators. The energy absorbed by the brake is dissipated in the form of heat. This heat will

dissipate into the surrounding atmosphere to stop the vehicle. Therefore, the braking system should have the following requirements:

The brakes must be strong enough to stop the vehicle at a minimum distance in an emergency.

a) The driver must correctly control the vehicle when braking, and the vehicle must not skid.

b) The brake must have good anti-fading performance, that is, its effectiveness should not be reduced under continuous long-term use.

c) The brake should have good anti-wear performance.

If there is no braking system, the vehicle will place passengers in an unsafe position. Therefore, all vehicles must have proper braking.

DESIGN OBJECTIVE AND OVERVIEW

The main purpose of the braking system is to convert the kinetic energy of the car into heat energy, thereby reducing the car's

performance. The brake system is designed as a hydraulic system, consisting of two cylinders, one that makes the first two tire brakes, and the other that has two rear tire brakes. Each master cylinder has two adjustable straps, one caliper per wheel, four system calipers, and four rotors or brake discs. The process of the braking system is as follows: the driver applies power to the brake pedal, and the brake station operates forcefully on the main cylinder, thus delivering the brake fluid to the master cylinder. After that, the extracted liquid is pressurized on each caliper, causing the caliper piston to apply gripping force to the rotor. Therefore, the input of the system is the foot force applied by the driver, and the output is the grip strength applied to the rotor by wire. Depending on the rules and regulations of the competition, the design of the brake system must be able to completely close all four-car tires in the event of an emergency brake. Therefore, in this type of compression design, the main purpose is to design a system so that the driver can apply a lot of energy but not excessively to the brake structure to completely lock the car tires. The system is designed for a foot capacity of approximately 2500N.

DEFINITION OF PROBLEM

The goal / specificity of the current task is to design and analyze the disc rotor made with SS410. SS materials are used to design the disc rotor. After that, the Rotor will be upgraded according to certain parameters to get the best design. The Rotor was created in Solid Works and submitted to ANSYS for analysis.

SELECTION OF COMPONENTS

A. Brake rotor and Calipers: The important requirement of the brake disc is:

- It should provide a surface with good anti-wear properties.
- It should allow the best heat transfer rate.
- Each cycle of brake application generates heat, which must be dissipated to the atmosphere.
- Sufficient strength and minimal weight. It must be contained in the smallest space available.

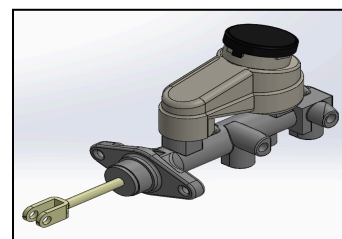
It is helpful to choose a rotor that is as wide as possible. This is because the distribution of the same torque, as the size increases, the proper power decreases. Therefore, a Rotor with a diameter of 160 mm is preferred for a 10-inch rim. A rotor is attached to promote heat transfer. The size of the disc is 3mm.

SPECIFICATION OF ROTOR

Material	SS410
Diameter	160 mm
PCD	80 mm
Thickness	3 mm

B. Master Cylinder:

Two master cylinders have been selected, to obtain two independent circuits for compression, and can be obtained with a single control of the brake base with a balance bar. It contains DOT3 as brake fluid. The circuit consists of solid pipes followed by flexible brake tubes connected to the drivers by Benjo bolts.



C. Brake Pedal:

The base of the brake is mechanically made from 6061 aluminum plate with two cylinders. Its design can withstand the force of 2000N on a ladder. The ratio of the pedal is set to 1:3.

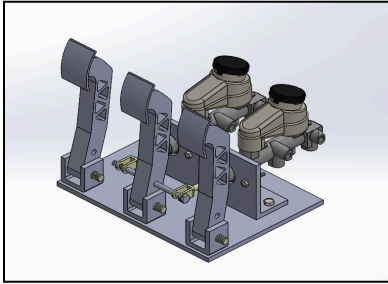


Fig: Pedal Assembly(CAD Module)

CALCULATIONS

STEP-1(STATIC WEIGHT DISTRIBUTION)

Total weight = 250 kg

Weight ratio= 45:55

Front weight = Total weight * 0.45

$$= 250 * 0.45$$

$$= 112.5 \text{ kg}$$

Rear weight = Total weight * 0.55

$$= 250 * 0.55$$

$$= 137.5 \text{ kg}$$

Static weight distribution

Front = $112.5 * 9.81$

$$= 1103.62 \text{ N}$$

Rear = $250 * 9.81$

$$= 1348.87 \text{ N}$$

STEP-2 (REQUIRED DECELERATION)

Assumed stopping distance = 9.37m

Assumed speed in Km/hr = 45 km/hr

Assumed speed in m/s = 12.5 m/s

Formula used

$$v^2 - u^2 = 2as$$

$$a = -u^2 / 2s \quad \{v=0\}$$

$$\text{Required deceleration}(d) = 12.5 * 12.5 / 16.66 = 8.33 \text{ m/s}^2$$

STEP-3 (WEIGHT TRANSFER)

Acceleration due to gravity(g) = 9.8 m/s^2

Wheel base (Wb) = 1.525 m

Weight (m) = 250 kg

Centre of gravity (Cg) = 0.3 m

Absolute weight transfer

$$\text{Weight transfer} = d * Cg * m / g * Wb$$

$$\text{Weight transfer} = 41.76 \text{ Kg}$$

STEP-4 (DYNAMIC WEIGHT TRANSFER)

Front = front weight + weight transfer

$$= 112.5 + 41.76$$

Front = 154.26 kg

Front = 1513.29 N

Rear = Rear weight – Weight transfer

$$= 137.5 - 41.76$$

Rear = 95.74 kg

Rear = 939.20 N

STEP-5 (BRAKING FORCE)

Coefficient of friction between Road and

Tyre = 0.7

F = COF x Dynamic Load

Braking Force

Front = Front weight transfer * COF

$$= 1513.29 * 0.7$$

$$= 1059.30 \text{ N}$$

Rear = Rear Weight transfer * COF

$$= 939.20 * 0.7$$

$$= 657.44 \text{ N}$$

Where, COF=Coefficient of Friction

STEP-6 (BRAKE TORQUE)

Theoretical braking torque can be calculated as follows

Radius of tire $R = 0.229 \text{ m}$

Torque = Force * Radius

Front = Front braking force * Radius of Tire
 $= 1059.30 * 0.229$
 $= 242.5 \text{ N/m}$

On each wheel = 182.84 N/m

Rear = Rear braking force * Radius of tire
 $= 657.44 * 0.229$
 $= 150.55 \text{ N/m}$

On each wheel = 75.275 N/m

APPLIED PARAMETERS

STEP-7 (PEDAL FORCE)

Pedal Ratio = 1:3

Pedal force = 500 N

Force on TMC = pedal ratio * pedal force
 $= 3 * 500$
 $= 1500 \text{ N}$

STEP-8 (PRESSURE CALCULATION)

Bore diameter (TMC) = 0.01905 m

Area of TMC = $\pi * \text{Bore diameter}^2 / 4$
 $= 3.14 * 0.01905^2 / 4$
 $= 2.84 \times 10^{-4} \text{ m}^2$

Pressure of TMC = Force of TMC / Area of TMC
 $= 1500 / 2.84 \times 10^{-4} = 5263.15 \text{ kPa}$

Where, TMC = Tandem Master Cylinder

PRESSURE OF TMC IS EQUAL TO PRESSURE OF CALLIPER (NO PRESSURE LOSS)

STEP-9 (CLAMPING FORCE)

CLAMPING FORCE FRONT

Bore diameter (calliper) = 0.034 m

No of pistons = 2

COF between pad and disc (μ) = 0.38

Area of calliper = $\pi * \text{Bore diameter}^2 / 4$
 $= 3.14 * 0.0292^2 / 4$
 $= 6.6 \times 10^{-4} \text{ m}^2$

Clamping force = No of pistons * COF between pad and disc (μ) * area of calliper * pressure of TMC
 $= 2 * 0.4 * 9.07 \times 10^{-4} * 5263.15 * 10^3$
 $= 3.81 \text{ kN}$

CLAMPING FORCE REAR

Bore diameter (calliper) = 0.034 m

No of pistons = 2

COF between pad and disc (μ) = 0.38 m²
Area of caliper = $\pi * \text{Bore diameter}^2 / 4$
 $= 4.9 \times 10^{-4} \text{ m}^2$

Clamping force = No of pistons * COF between pad and disc (μ) * area of caliper * pressure of TMC
 $= 2 * 0.4 * 9.07 \times 10^{-4} * 5263.15 * 10^3 = 3.81 \text{ kN}$

STEP-10 (braking torque)

Effective radius = $D_3 - d_3 / 3 * D_2 - d_2$
 $= 111.20 \text{ mm}$

Torque $T = \text{Force} * \text{Effective Radius}$
 $= 3.81 * 0.1112$
 $= 422.9 \text{ N/m}$
 $= 123.64 \text{ N/m}$

Where, Effective radius = the effective radius of the disc (measured from the disc centre of rotation to the mean centre of pressure of the calliper pistons)

FINITE ELEMENT ANALYSIS

A. Static structural analysis:

In this project, structural and thermal analysis of two disc rotors, i.e., drilled disc rotor and a slotted disc rotor are performed to compare the results and derive inferences and conclusions. However, the same methodology followed for the analysis of both the rotor. Steady state thermal analysis coupled with static structural analysis was performed in Ansys. The CAD model of rotor was created in Solidworks then was exported for analysis.

B. Transient Thermal Analysis:

Temperature changes in rotor geometry are calculated and analyzed here. As the brakes are applied to the disc brake rotor of the car, the friction will produce heat, so the temperature generated must be driven and spread across the cross section of the disc rotor. Braking conditions are very difficult, so a hot analysis should be done. Temporary thermal analysis determines the distribution of temperature under conditions of high load.

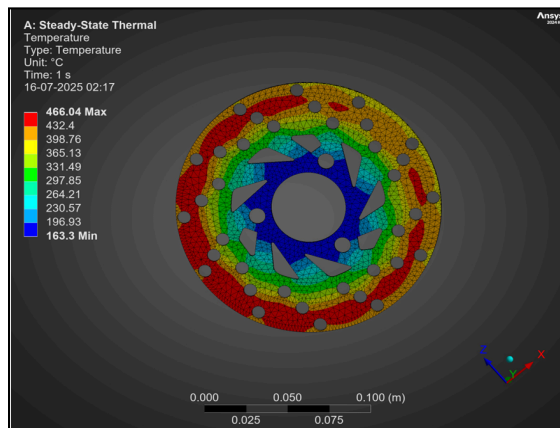


Fig : - Brake Rotor Thermal Analysis

C. Brake Pedal Analysis: The brake pedal converts driver-applied force

into hydraulic pressure for braking.

Analysis ensures the pedal can withstand peak loads without deformation, maintaining stiffness and safety. Stress distribution, deformation, and factor of safety are evaluated to guarantee durability, responsiveness, and reliable performance under repeated high-load conditions.

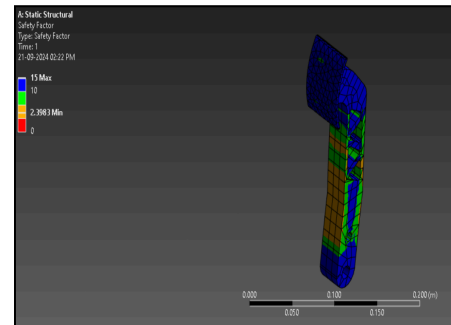


Fig:Brake Pedal Analysis

Conclusion:

A 160 mm diameter, 3 mm thick rotated and molded disc rotor was designed in SolidWorks and analyzed in ANSYS. Results (Table VI) show both rotors are safe for construction, with total pressures within SS 410 limits. Deformations were 0.027101 mm for the rotated rotor and 0.016668 mm for the molded rotor. Temperature changes were similar for both, but the molded rotor exhibited slightly lower total and ambient temperatures, attributed to increased thermal conductivity from drilling during braking. This indicates that the molded rotor offers improved heat dissipation while maintaining structural safety under operating conditions.